

## Answer Key — Core Mathematics Grade 10

Sub-Strand 2.4: Trigonometry 1 • Lesson 7: Angles of Elevation and Depression: Solving Real Height-and-Distance Problems Across Kenya

Teacher reference only. Do not project or hand out.

### Quick marking grid

| Q1 | Q2 | Q3 | Q4 | Q5 | Q6 | Q7 |
|----|----|----|----|----|----|----|
| D  | A  | B  | A  | B  | C  | D  |

USE AFTER: PREDICT PHASE — Predict: Think-Pair-Share on the Nakuru mast (40 m, about  $50^\circ$ )

**Q1. A student stands 40 m from the base of a mast and measures the angle of elevation to the top as  $50^\circ$ . Let  $h$  be the height of the mast in metres. The 40 m side is next to the  $50^\circ$  angle, and  $h$  is opposite it. Which equation should be used to find  $h$ ?**

- A.  $\sin(50^\circ) = h/40$
- B.  $\tan(50^\circ) = 40/h$
- C.  $\cos(50^\circ) = h/40$
- D.  $\tan(50^\circ) = h/40$**

**Correct answer: D.** The known side (40 m) is adjacent to the angle and the unknown height is opposite it. The ratio that links opposite and adjacent is tangent, so  $\tan(50^\circ) = h/40$ .

USE AFTER: OBSERVE PHASE — Part A: Diagram Construction (elevation and depression diagrams)

**Q2. Which statement correctly describes an angle of depression?**

- A. The angle measured downward from the horizontal to the line of sight**
- B. The angle between the ground and the observer's feet
- C. The angle measured between the line of sight and the vertical line through the observer
- D. The angle measured upward from the horizontal to the line of sight

**Correct answer: A.** An angle of depression is measured downward from the observer's horizontal line to the line of sight to an object below. The upward version is the angle of elevation.

USE AFTER: OBSERVE PHASE — Part B: Worked Demonstration (Fort Jesus cliff depression example, 20 m and  $35^\circ$ )

**Q3. A person stands at the top of the Fort Jesus cliff in Mombasa, 20 m above sea level. The angle of depression to a dhow at sea level is  $35^\circ$ . What is the horizontal distance from the base of the cliff to the dhow, to 1 decimal place?**

- A. 34.9 m
- B. 28.6 m**
- C. 24.4 m
- D. 14.0 m

**Correct answer: B.** The cliff height (20 m) is opposite the  $35^\circ$  angle and the distance  $d$  is adjacent.  $\tan(35^\circ) = 20/d$ , so  $d = 20 \div \tan(35^\circ) = 20 \div 0.7002 \approx 28.6$  m.

USE AFTER: EXPLAIN PHASE — Problem 1 (Elevation — Nairobi): flagpole in Uhuru Park

**Q4. A visitor at Uhuru Park, Nairobi stands 15 m from the base of a flagpole. The angle of elevation to the top of the flagpole is  $62^\circ$ . What is the height of the flagpole, to 1 decimal place?**

- A. 28.2 m**
- B. 13.2 m
- C. 8.0 m
- D. 7.0 m

**Correct answer: A.** The distance of 15 m is adjacent and the height is opposite, so  $h = 15 \times \tan(62^\circ) = 15 \times 1.8807 \approx 28.2$  m.

USE AFTER: EXPLAIN PHASE — Problem 2 (Elevation — Kisumu): surveyor and five-storey building

**Q5. A surveyor in Kisumu stands 42 m (horizontal distance) from a five-storey building. The angle of elevation to the top of the building is  $38^\circ$ . What is the height of the building, to 1 decimal place?**

- A. 53.8 m
- B. 32.8 m**
- C. 25.9 m
- D. 33.1 m

**Correct answer: B.** The 42 m is adjacent to the angle and the height is opposite, so  $h = 42 \times \tan(38^\circ) = 42 \times 0.7813 \approx 32.8$  m.

USE AFTER: EXPLAIN PHASE — Problem 3 (Depression — Mombasa): tourist at Fort Jesus cliff and dhow

**Q6. A tourist stands at the top of a cliff 27 m above sea level. She sees a dhow at sea level at an angle of depression of  $41^\circ$ . Let  $d$  be the horizontal distance from the base of the cliff to the dhow. Which working correctly finds  $d$ ?**

- A.  $d = 27 \div \sin(41^\circ) \approx 41.2$  m
- B.  $d = 27 \times \tan(41^\circ) \approx 23.5$  m
- C.  $d = 27 \div \tan(41^\circ) \approx 31.1$  m**
- D.  $d = 27 \times \cos(41^\circ) \approx 20.4$  m

**Correct answer: C.** The cliff height (27 m) is opposite the  $41^\circ$  angle and  $d$  is adjacent.  $\tan(41^\circ) = 27/d$ , so  $d = 27 \div \tan(41^\circ) = 27 \div 0.8693 \approx 31.1$  m. Writing  $\tan(41^\circ) = d/27$  is the common error.

USE AFTER: MODEL BUILDING PHASE — Cumulative right-triangle model: depression inset labelled 'Angle of Depression = Angle of Elevation (alternate interior angles)'

**Q7. From the top of a cliff, the angle of depression to a boat on the sea is  $41^\circ$ . What is the angle of elevation from the boat to the person at the top of the cliff?**

- A.  $139^\circ$
- B.  $90^\circ$
- C.  $49^\circ$
- D.  $41^\circ$**

**Correct answer: D.** The horizontal line at the cliff top and the horizontal line at sea level are parallel. Alternate interior angles are equal, so the angle of elevation from the boat equals the angle of depression from the cliff top, which is  $41^\circ$ .